

PoP-ECC:

Robust and Flexible Error Correction against Multi-Bit Upsets in DNN Accelerators

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Background & Motivation



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DNNs and Errors

Two types of errors

Data errors:

Small noises on the input data

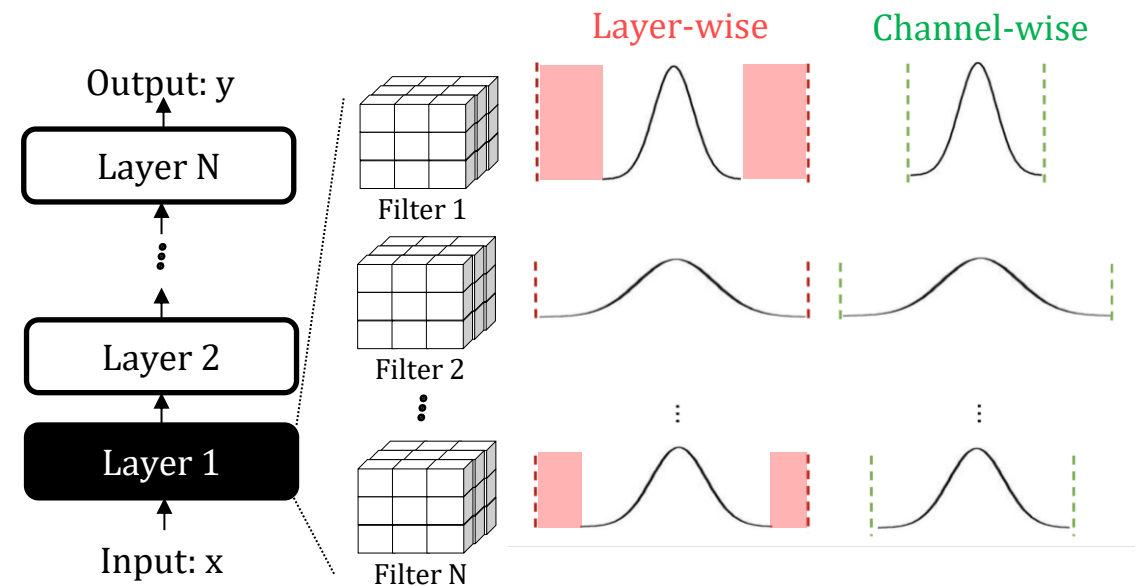
Bit errors:

Random bit flips during storage

DNNs and Data Errors

- DNNs inherently tolerate small data errors, facilitating quantization.
- Quantization schemes
 - Layer-wise vs. channel-wise
 - Channel-wise quantization provide better accuracy.

Model	FP32	8-bit Quantization	
		Layer	Channel
ResNet50	76.13%	-0.12%	-0.03%
DenseNet169	75.60%	-0.08%	-0.00%
Inception V3	77.29%	-0.40%	-0.11%
MobileNet V2	71.88%	-1.66%	-0.79%
EfficientNet-B2	80.61%	-0.17%	-0.03%
RegNetX-8GF	79.34%	-0.05%	+0.04%
Average		-0.41%	-0.15%

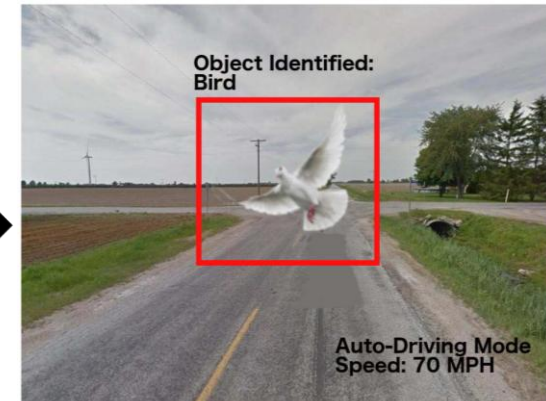
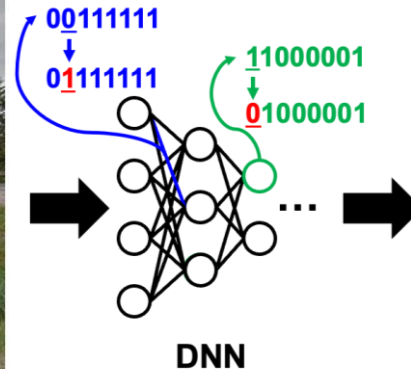
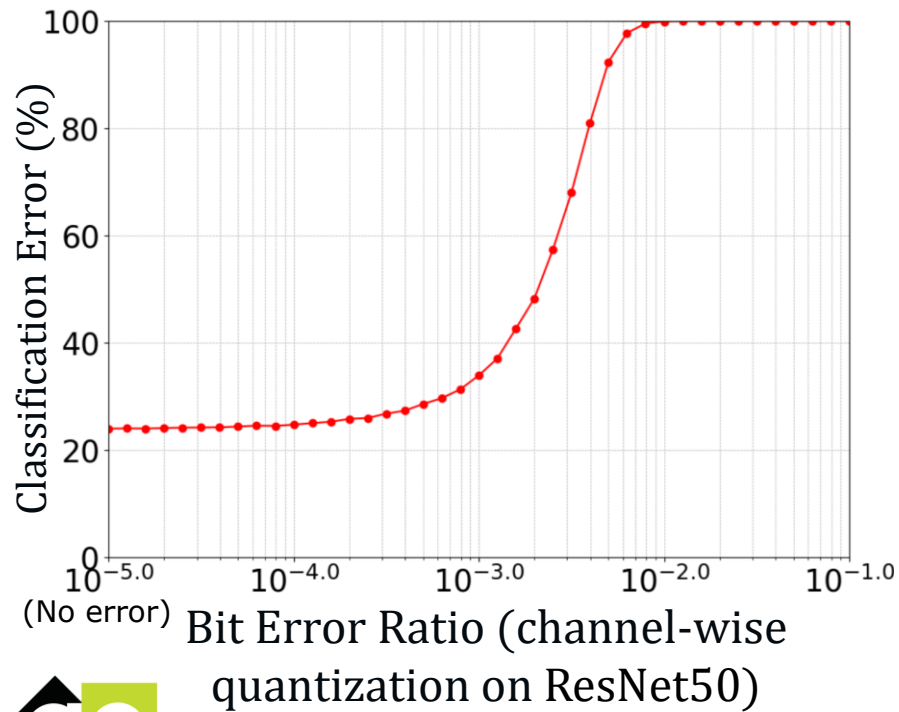


A comparison of accuracy degradation between layer- and channel-wise quantization (uniform quantization)

A. Gholami, S. Kim, Z. Dong, Z. Yao, M. W. Mahoney, and K. Keutzer, "A survey of quantization methods for efficient neural network inference," 2021. [Online]. Available: <https://arxiv.org/abs/2103.13630>

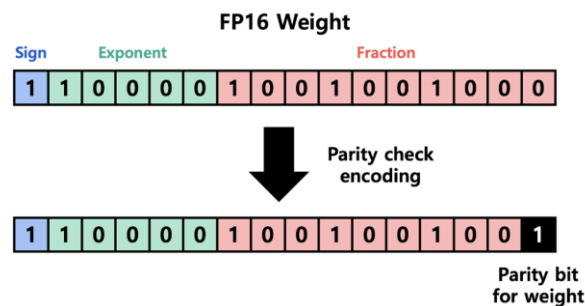
DNNs and Bit Errors

- **Bit errors on most significant bits** affect DNN accuracy by changing the data value significantly.

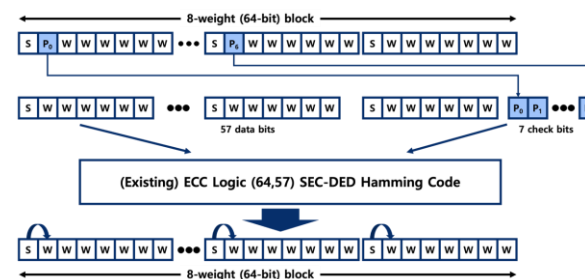


Prior Work on SRAM Protection

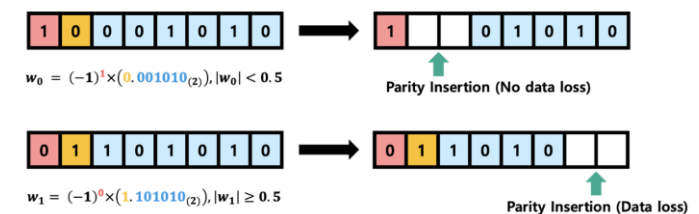
- Many DNN processors have large **SRAMs** to store coefficients and activations.
- Prior work on SRAM protection:
 - Weight Nulling [1]
 - In-place Zero-space ECC [2]
 - Value-aware Parity Insertion ECC [3]



Weight Nulling



In-place Zero-space

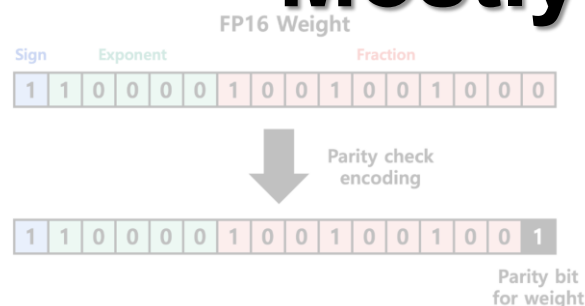


Value-aware Parity Insertion ECC

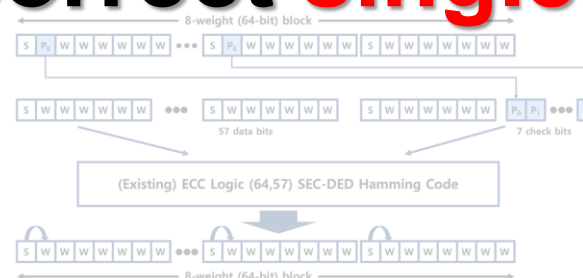
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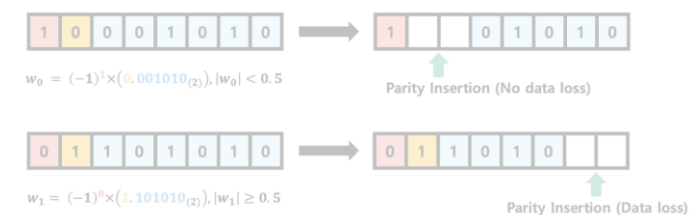
Mostly can correct single-bit errors!



Weight Nulling



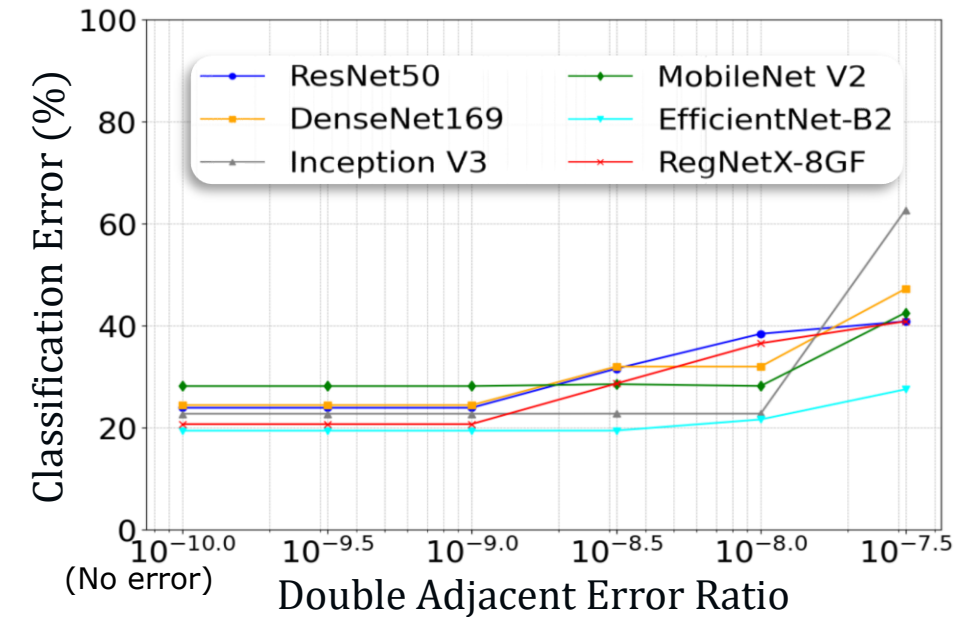
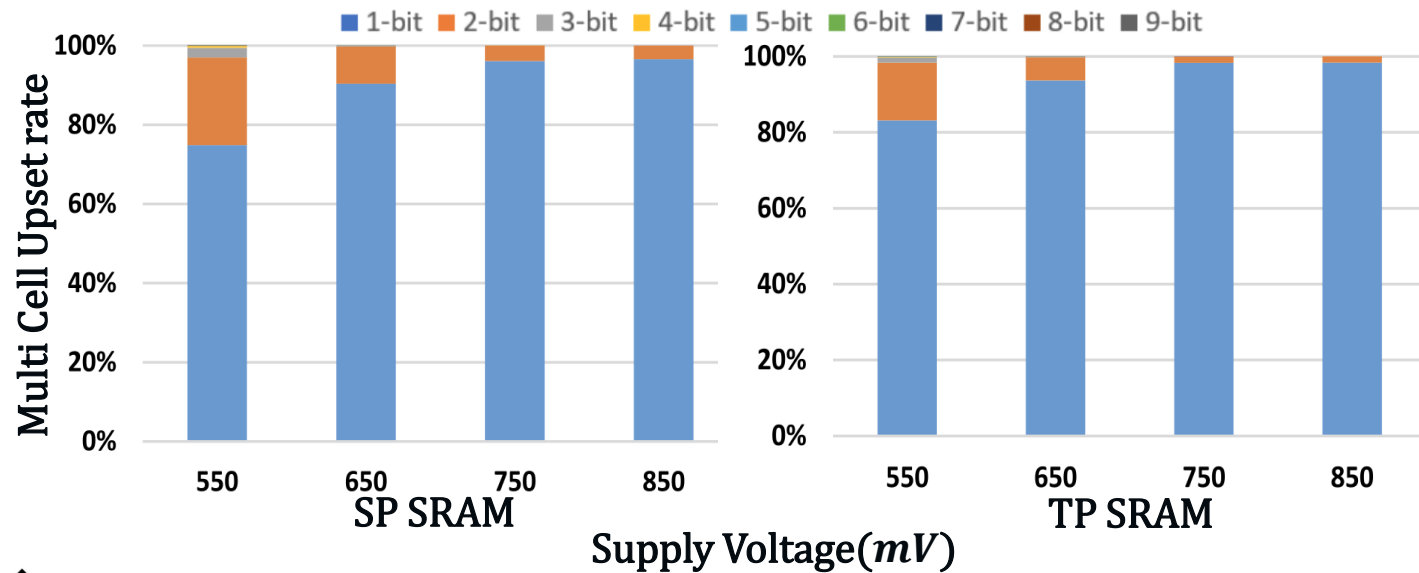
In-place Zero-space



Value-aware Parity Insertion ECC

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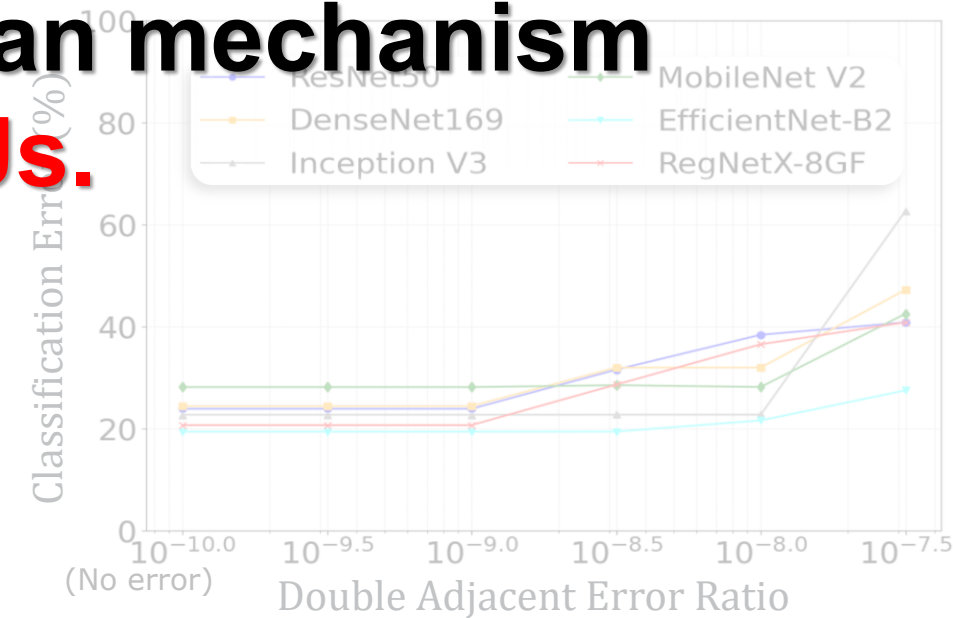
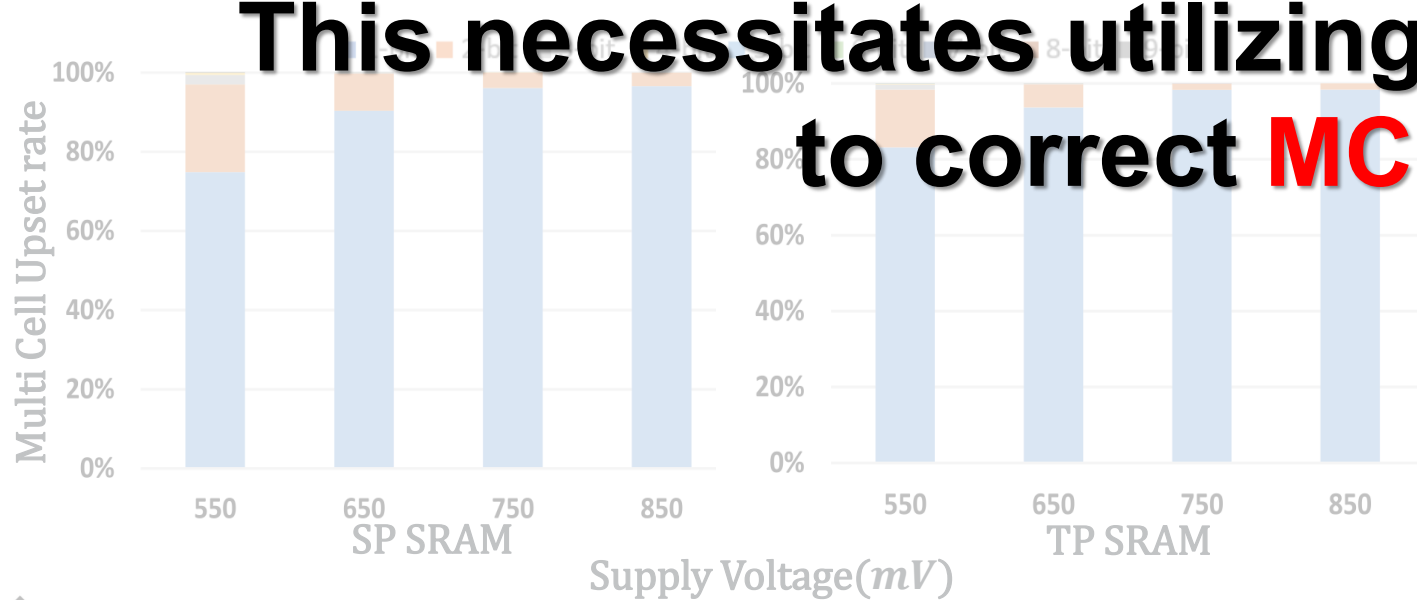
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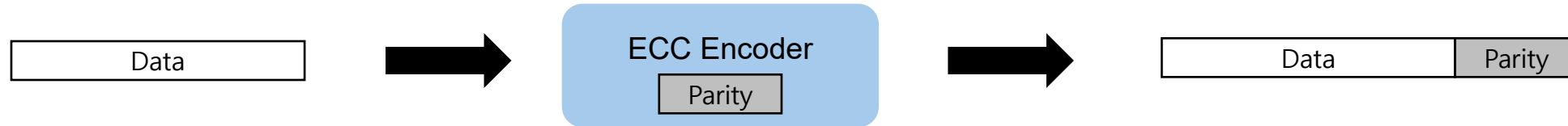
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This necessitates utilizing an mechanism to correct MCUs.



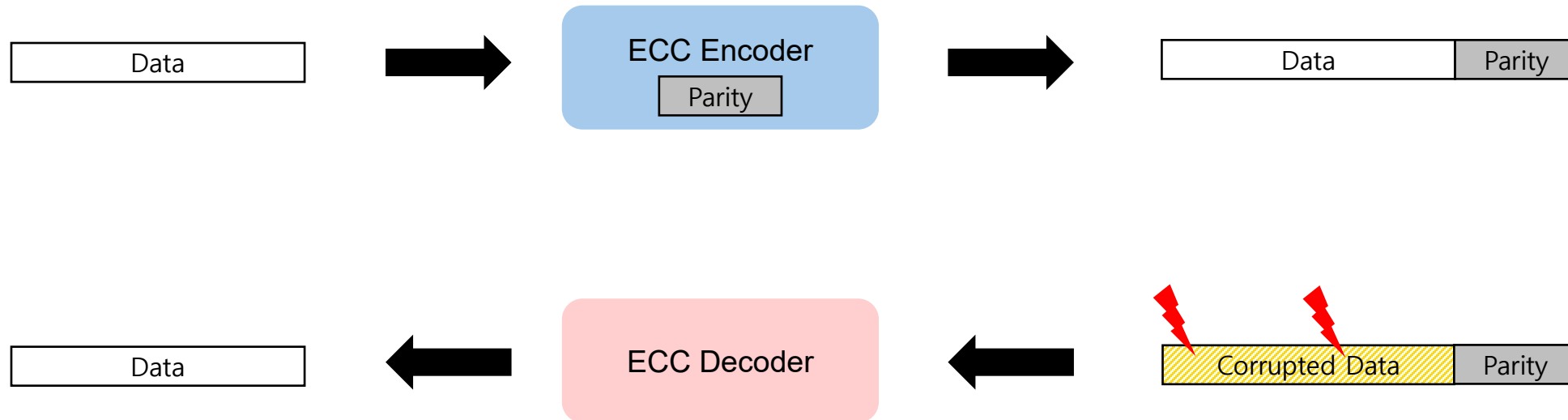
Error Correction Codes (ECC)

- A common solution to DNN reliability is using **ECC**.



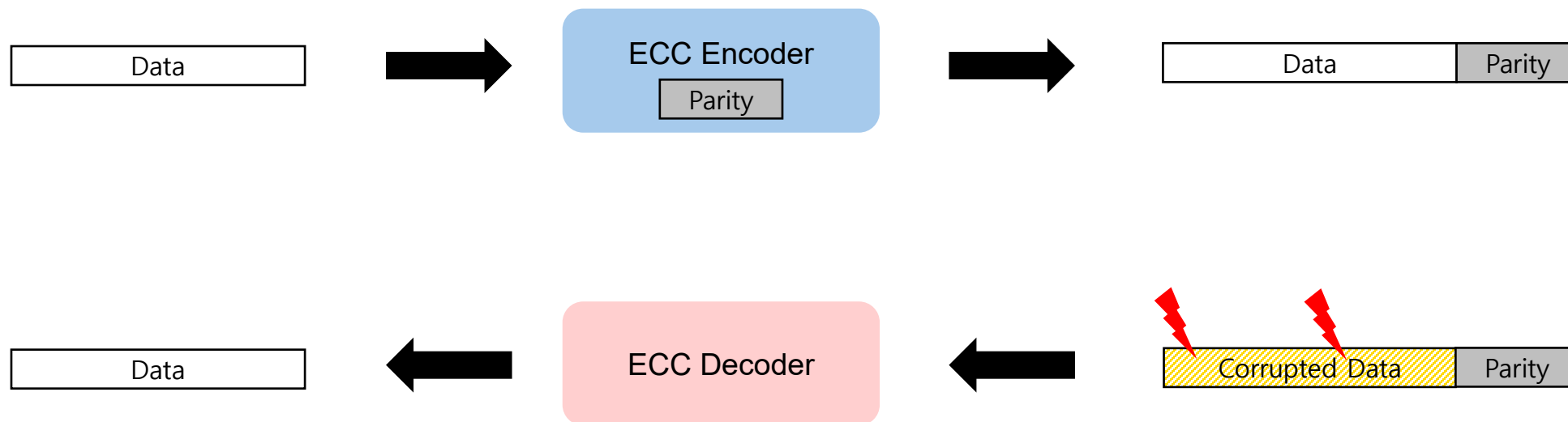
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Parities of Parities ECC (PoP-ECC)

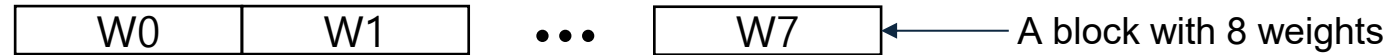


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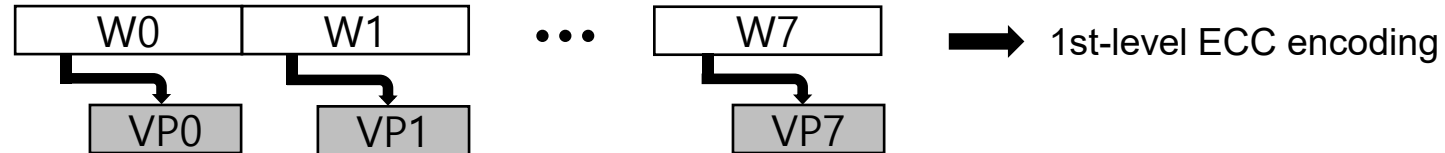
PoP-ECC

- PoP-ECC is a **strong** and **flexible** two-level ECC scheme to prevent severe memory errors.
 - First level:** extracts **Virtual Parities (VPs)** from the weights.



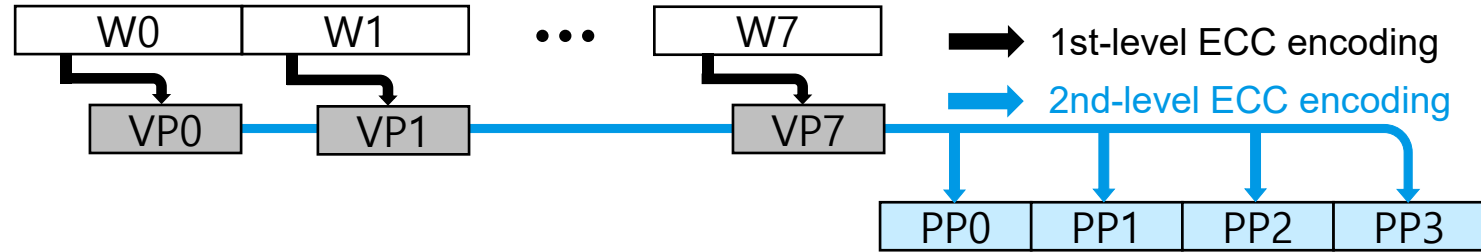
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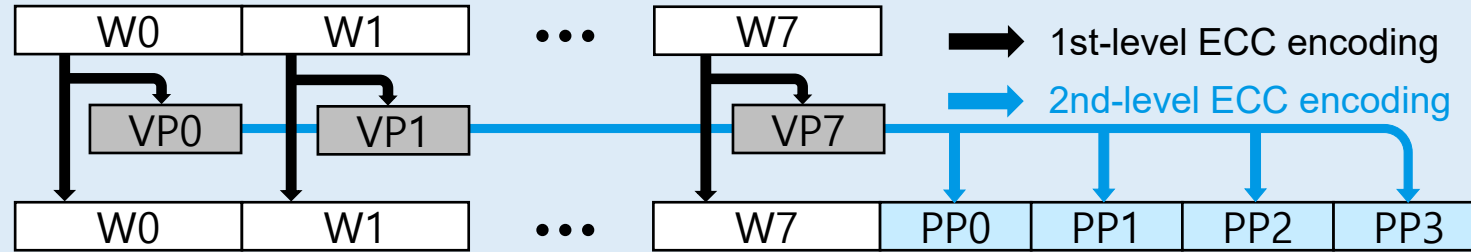
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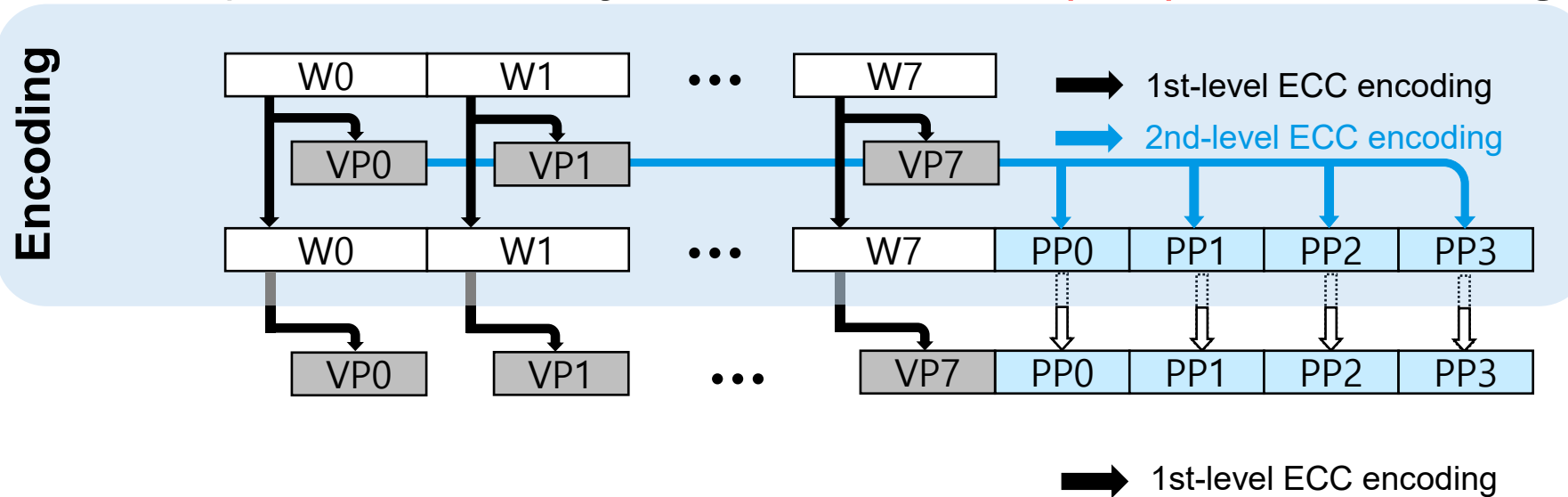
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Encoding



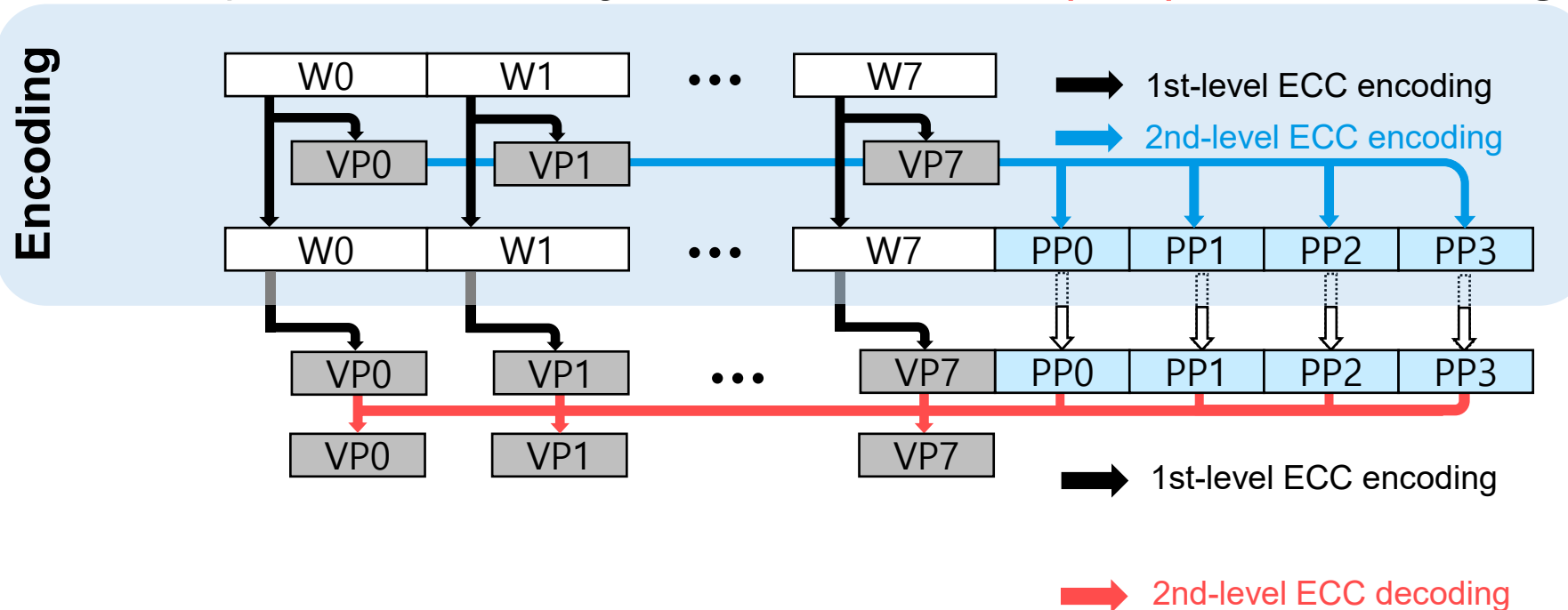
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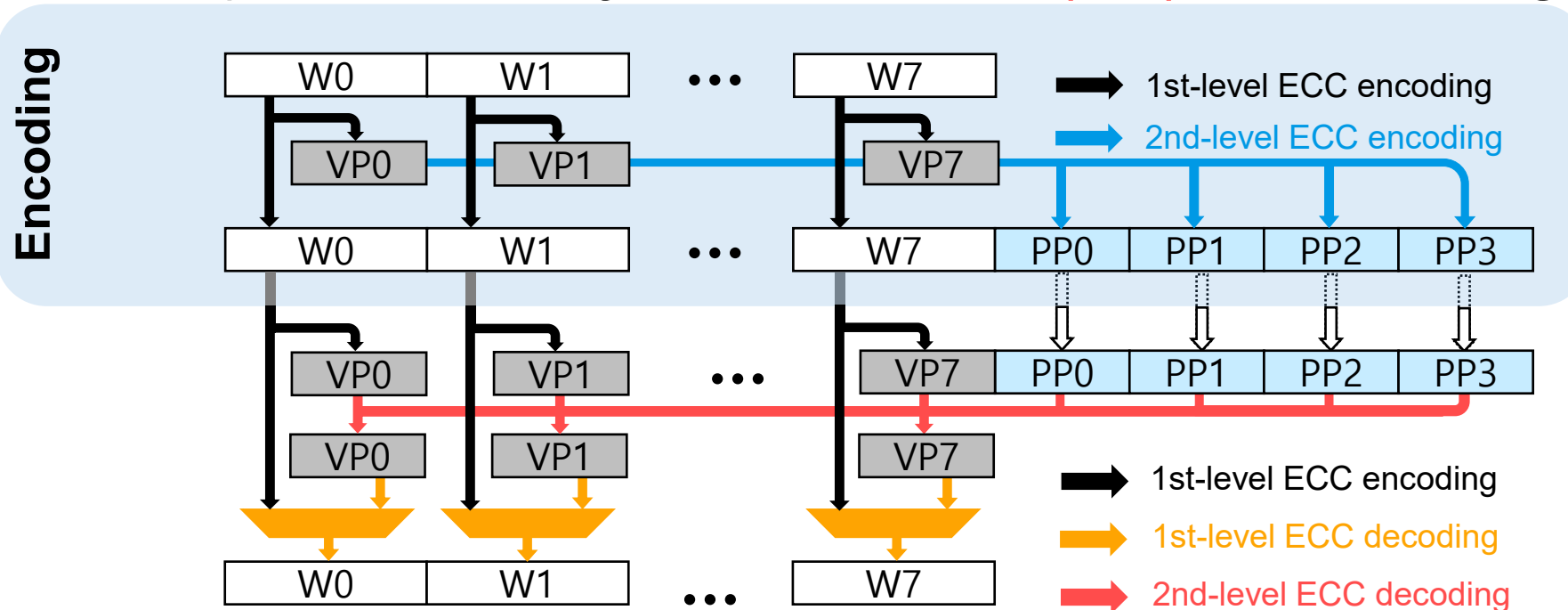
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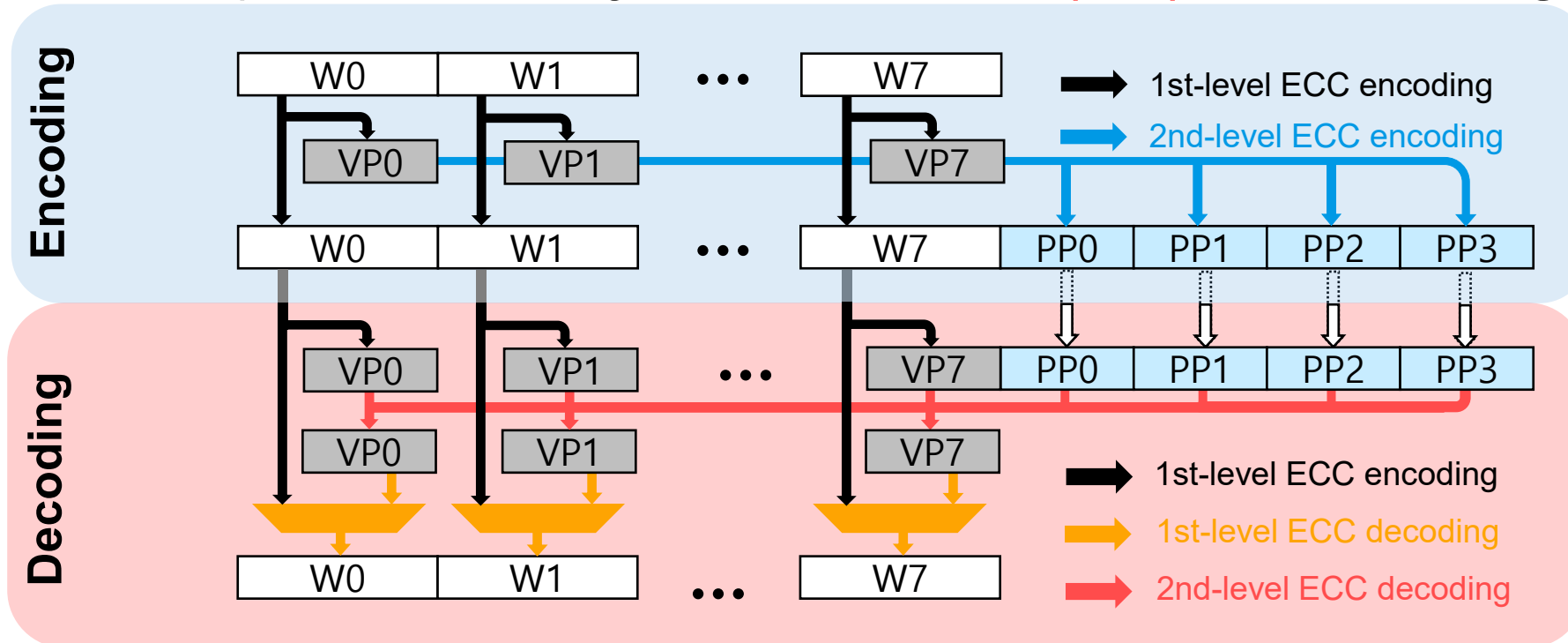
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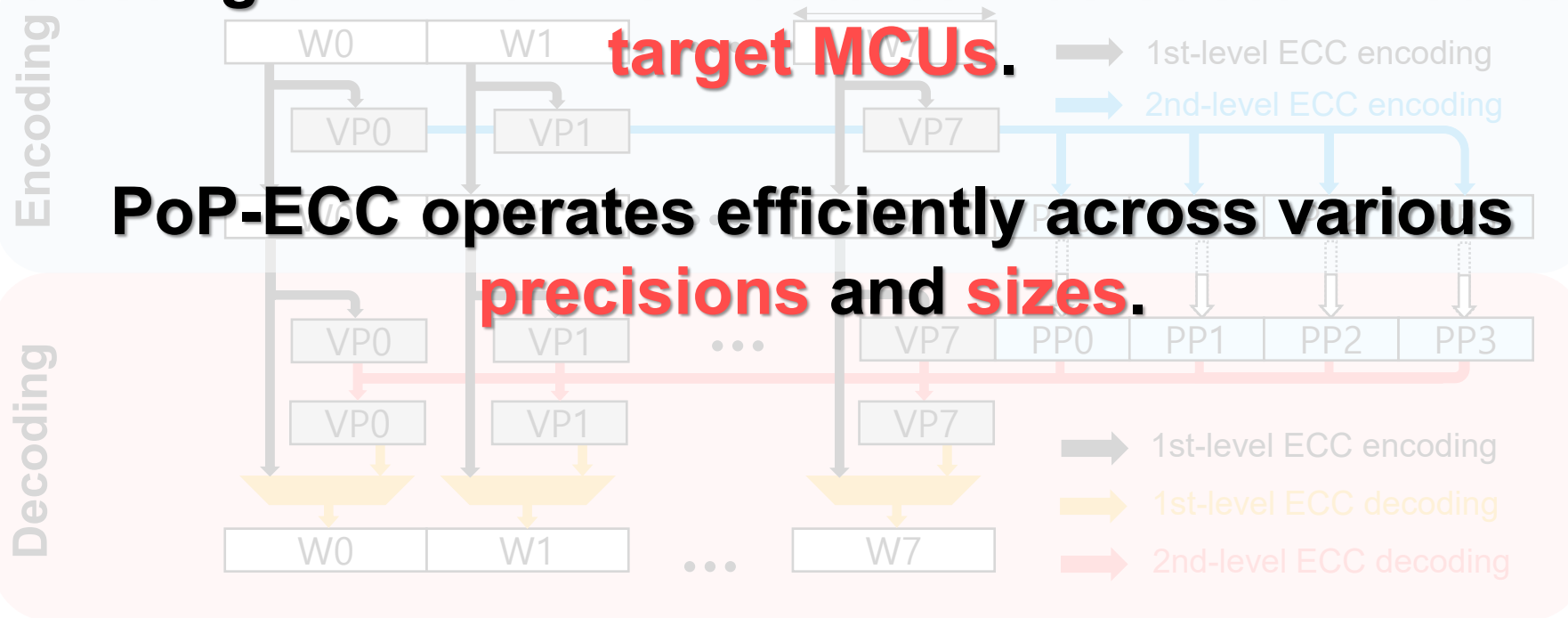
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Extracting a VP from raw data can be customized to the target MCUs.



Quantization & PoP-ECC (Q+PoP)

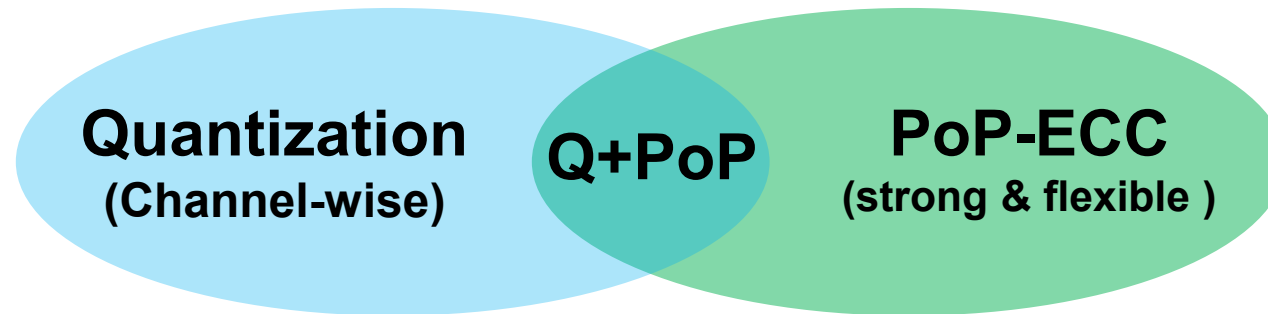


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Q+PoP

- **Q+PoP** combines the **high accuracy** of channel-wise quantization with the **robustness** of PoP-ECC without extra SRAM for ECC redundancy.



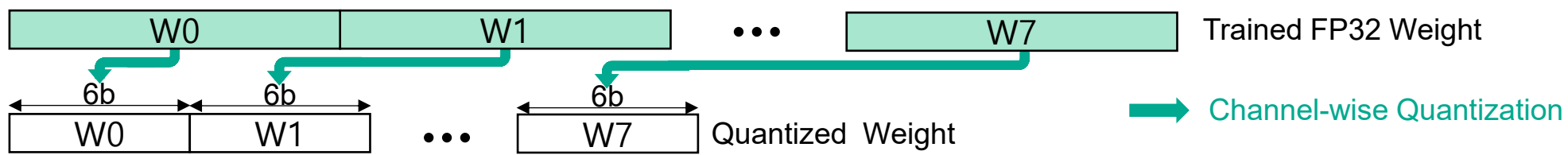
Q+PoP encoding process

- Without loss of generality, **Q+PoP** reduces FP32 to INT6 using channel-wise quantization.



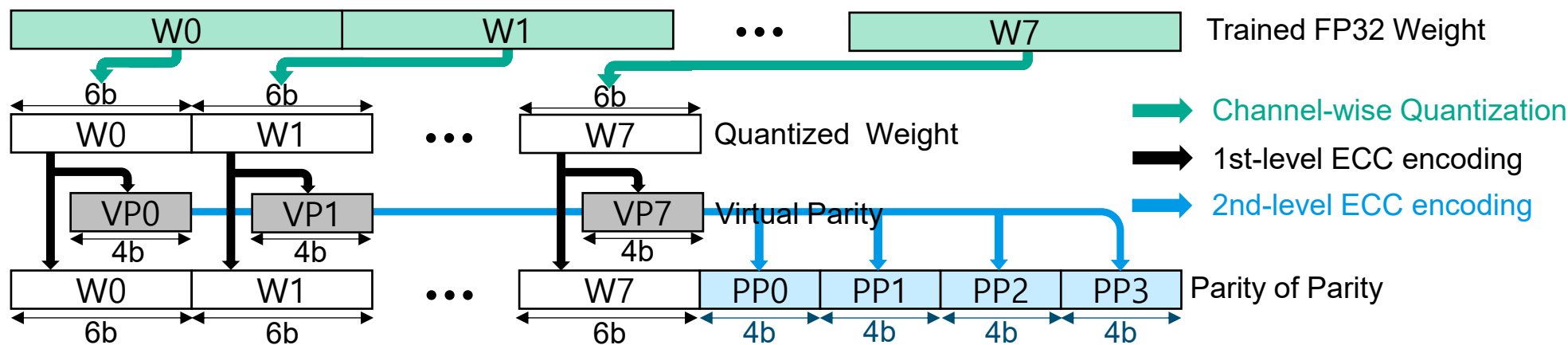
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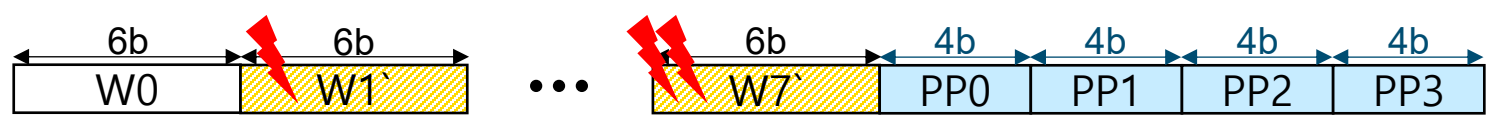
Q+PoP encoding process

- In the first stage, a **4-bit Virtual Parity (VP)** is generated from the **6-bit weight**.
 - The generated 4-bit VP is designed to correct all single- and double-bit errors in the upper 5-bits of the quantized weight.
- In the second stage, PoP-ECC utilizes RS code to protect VPs on a symbol basis.
 - Q+PoP applies **4-bit RS encoding** to the eight VPs to **generate 4 PPs**.



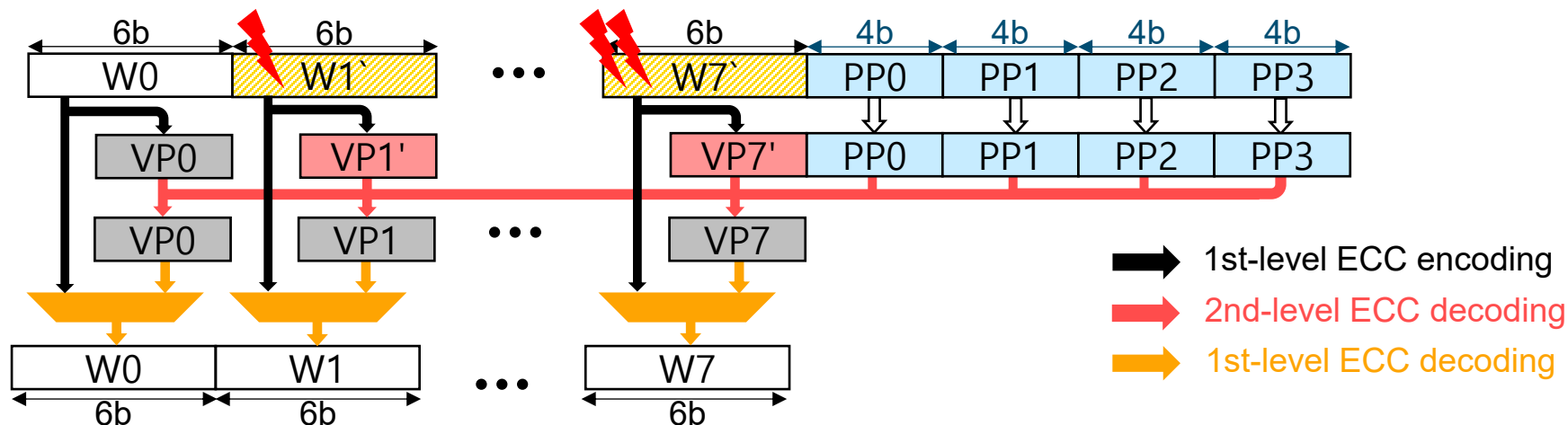
Q+PoP decoding process

- On receiving the **erroneous block**, PoP-ECC first regenerates VPs using the same generator matrix as the encoding.



Q+PoP decoding process

- **(12, 8) RS code** decoding can correct **two erroneous 4-bit VPs** with four **4-bit PPs**.
- After restoring the VPs, PoP-ECC utilizes the VPs to restore the weights.



Evaluation



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Evaluation

- We implemented *WN*, *VAPI*, and **Q+PoP** in the **PyTorch** framework.

Methodology				
Mem. protection schemes	DNN Models	Error Injection	Input dataset	Error ratios
WN	ResNet50	SE (Single Error)	ISLVR-2012	10 ⁻⁵ to 10 ⁻¹
VAPI	DesneNet169	DAE (Double	(50,000 validation	
Q+PoP	Inception V3	Adjacent Error)	images)	
	MobileNet V2			
	EfficientNet-B2			
	RegNetX-8GF			

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Accuracy

- Accuracy without errors (shows the impact of quantization)

Reliability

- Accuracy with errors

Hardware overheads



Base Top-1 classification Accuracy

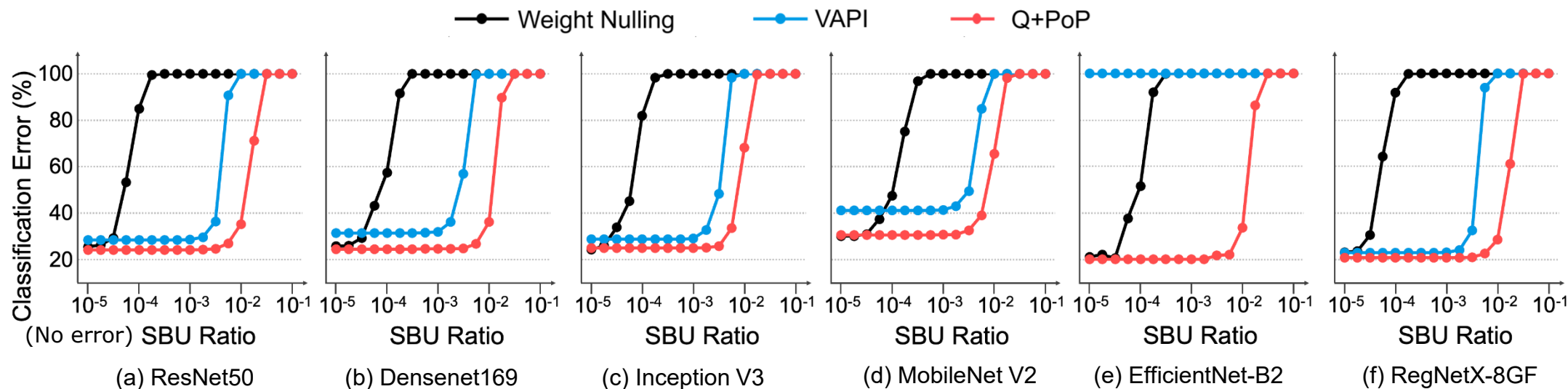
- **Q+PoP** shows an average of **−1.07%** degradation.

Model	FP32	WN	VAPI	Q+PoP
ResNet50	76.13%	−0.00%	−4.25%	−0.41%
DenseNet169	75.60%	+0.02%	−7.05%	−0.24%
Inception V3	77.29%	−0.01%	−4.89%	−2.35%
MobileNet V2	71.88%	−0.01%	−16.07%	−2.47%
EfficientNet-B2	80.61%	−0.00%	−80.51%	−0.78%
RegNetX-8GF	79.34%	+0.01%	−2.01%	−0.17%
Average		0.00%	−19.13%	−1.07%

- WN provides almost the same accuracy as the pretrained FP32 weights.
- VAPI sacrifices accuracy significantly (up to 80.51%) in our evaluation.

Reliability against Single-Errors

- **Q+PoP** can provide **higher accuracy** than **VAPI**, for lower SBUs ($< 10^{-5}$).
- **Q+PoP** can tolerate the highest BER of up to near 3×10^{-3} .

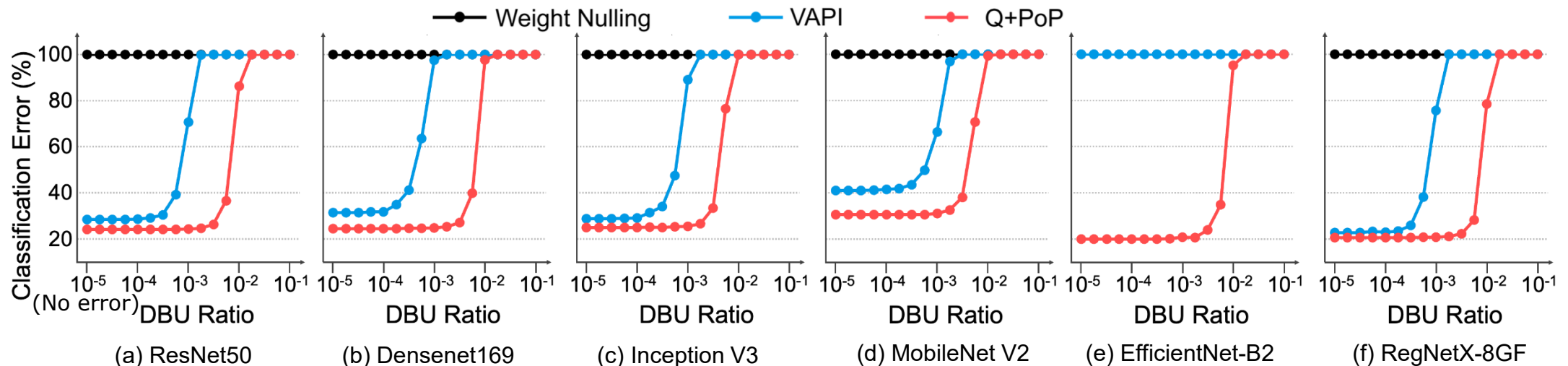


- The accuracy of WN decreases quickly as the ratio approaches 10^{-4} .

Reliability against Double-Adjacent Errors

- **Q+PoP** can tolerate:

- up to 10^{-3} by correcting up to two DAEs within a 64-bit block.
- up to **31.62x higher *maximum DAE** ratio than VAPI.



* Maximum DAE ratio is that which does not reduce DNN accuracy by 0.1 percentage point than error-free cases.



- **WN cannot detect double-bit errors** within a weight even at lower DBUs ($<10^{-5}$).
- **At 10^{-3} , VAPI cannot protect DNN accuracy.**

Hardware Costs

- Synthesis Set-up
 - Compiler : Synopsys design compiler
 - Logic libraries : UMC 28nm SVT library.

Hardware Cost	Speed	Area	Power
PoP-ECC Encoder	0.65ns	92um ² (0.3K NAND2 gates)	0.08mW
PoP-ECC Decoder	1.56ns	1760um ² (6K NAND2 gates)	1.2mW

Conclusion



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Technology scaling => Rising **soft error rates**
“**Multi-Cell Upsets (MCUs)** now common”

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Our Solution – **PoP-ECC**
Robust and flexible Two-level ECC: **Virtual Parities + Parities of Parities**
Effectively mitigates MCUs with **minimal overhead**

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Tailored for efficient DNN protection

Up to 31.62× better DAE tolerance
Higher accuracy with negligible delay, area, and power cost



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Michael B. Sullivan

NVIDIA



AI



Security



Systems



EDA



Design



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CONFERENCE

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